# SILAS KWABLA GAH

PhD Candidate in Computer Vision · 3D Geometric Perception, Reliability & Foundation Models

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Research Profile

Final-year PhD candidate in Computer Vision and Machine Learning at the University of Ghana (thesis defence scheduled for October 2026). Research investigates **reliable 3D geometric perception**, **inference-time foundation-model coordination** ($\Delta\theta = 0$), **probabilistic calibration**, and **process-level self-supervision**. Methodological focus lies on non-parametric arbitration across heterogeneous visual encoders, mitigating premature semantic collapse via compact candidate retention, 2D-to-3D geometric lifting, and cross-view physical consistency. Experienced with large-scale point-cloud benchmarks (ScanNet200, ScanNet++), PyTorch GPU acceleration, and streaming evaluation pipelines. Strongly aligned with the **CVI^2** research agenda at the Interdisciplinary Centre for Security, Reliability and Trust (**SnT**) in trustworthy computer vision, 3D shape understanding, multi-sensor perception, and visual generative modeling.

Education

PhD in Computer Vision and Machine Learning — **Expected defence: October 2026**
University of Ghana, Legon — *Supervisor: Prof. Ebenezer Owusu*

Thesis: *Visual Perception under Low Supervision: Inference-Time Coordination of Frozen Foundation Model Representations*

Research investigates frozen foundation-model composition, open-vocabulary 3D/2D segmentation, semantic evidence preservation, multi-view geometric consensus, and reliability-calibrated inference.

MPhil in Computer Science — **2016**
University of Ghana, Legon — *Accra, Ghana*

BSc in Information and Communications Technology — **Second Class Honours (Upper Division) · 2012**
Ghana Institute of Management and Public Administration (GIMPA) — *Accra, Ghana*

HND in Electrical & Electronic Engineering — **2009**
Accra Technical University — *Accra, Ghana*

Doctoral Research Highlights & Systems Rigor

Reliable 3D Point-Cloud Perception & Foundation-Model Composition (ACCV 2026, WACV 2027, ICLR 2027)

- Developed training-free open-vocabulary 3D point-cloud segmentation coordinating dense 3D vision-language representations (RegionPLC) with 2D promptable segmentation models (SAM 2.1) via multi-view geometric consensus.
- Addressed the generalized few-shot 3D benchmark on **ScanNet200** and **ScanNet++** (312 indoor scenes, ~50M points), establishing new state-of-the-art harmonic mean (34.87 novel mIoU, +6.40 points improvement) with rigorous paired bootstrap 95% confidence intervals [5.24, 7.64].
- Discovered and characterized **premature semantic collapse** under hard argmax operations in multi-model pipelines; proved that preserving top-k candidate distributions prevents irrevocable hypothesis loss, enabling downstream geometric arbitration to recover overlooked categories (formalized in an **ICLR 2027** submission).
- Designed streaming, memory-bounded evaluation pipelines utilizing float16 caching, spatial coordinate transforms, and chunked processing to evaluate dense per-point distributions on memory-constrained GPUs.

Reliability-Calibrated Adaptive Semantic Arbitration (RCASA) (Target: IEEE TPAMI In Prep)

- Formulated **RCASA**, a principled probabilistic framework coordinating heterogeneous visual models at inference time without gradient updates ($\Delta\theta = 0$), uniting Platt confidence calibration, Bernoulli uncertainty modeling, and contextual reliability weighting.
- Proved class-wise bounded updates and probability simplex preservation, providing formal trust guarantees for model composition under distribution shift.

Multi-View Consistency as Label-Free Process Supervision (SACAIR 2026 Submission)

- Formulated physical cross-view agreement between 2D images and 3D geometry as an intrinsic, label-free process reward for multimodal reasoning models (**Qwen2.5-VL-7B**), eliminating reliance on subjective human reward labels.
- Built an end-to-end online **GRPO/VERL** training pipeline on **vLLM**; conducted counterfactual stress-testing across 1,000+ rollouts, mitigating reward-hacking and reducing spatial area bias correlation ($\rho > 0.72 \rightarrow 0.334$).

Selected Publications and Manuscripts

- **Beyond Argmax: A Mechanistic Study of Semantic Retention in Frozen Foundation-Model Composition for Generalized Few-Shot 3D Segmentation.**
  International Conference on Learning Representations (ICLR 2027 Conference Submission; Preprint Available). First Author.
- **Training-Free Open-Vocabulary 3D Point-Cloud Segmentation on the Generalized Few-Shot Benchmark.**
  Asian Conference on Computer Vision (ACCV 2026, Paper #482; Preprint Available). First Author.
- **Training-Free Generalised Few-Shot Segmentation through Open-Vocabulary Semantic Arbitration.**
  IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2027, Paper #167). First Author.
- **Cross-View Agreement as a Label-Free Process Reward for Multimodal Reasoning.**
  Southern African Conference for AI Research (SACAIR 2026 Submission). First Author.
- **Transfer Learning-Based Encoder–Decoder Model for Skin Lesion Segmentation.**
  Communications in Computer and Information Science (CCIS), Vol. 1841, pp. 117–128, Springer, 2023. Co-Author.
- **Reliability-Calibrated Adaptive Semantic Arbitration of Frozen Foundation Model Representations.**
  Manuscript in preparation (targeted for IEEE TPAMI). First Author.

Technical Skills

Programming	Python, Java, C++
Deep Learning & RL	PyTorch, Transformers, vLLM, LoRA/QLoRA, SFT, GRPO, VERL
3D & Geometric Vision	Point clouds, camera intrinsics/extrinsics, coordinate transformations, 2D–3D lifting, multi-view geometric consensus, ScanNet200/ScanNet++
Foundation Models	RegionPLC, SAM 2.1, GroundingDINO, CLIP, DINOv2, Qwen2.5-VL
Trustworthy AI / Stats	Confidence calibration (Platt scaling, ECE, Brier score), uncertainty quantification, paired bootstrap confidence intervals, counterfactual validation
GPU & Infrastructure	CUDA fundamentals, Triton kernel benchmarking, Linux/HPC environments, streaming pipelines

Teaching, Professional Experience & Academic Service

Part-time Lecturer — Accra Technical University (2022 – Present)

- Lecturing in computing, algorithms, and applied machine learning; previously taught at GIMPA.

Founder & Lead Architect — EED Soft Consult (2018 – Present)

- Architected and delivered large-scale, production-grade distributed and mobile systems for enterprise and public-sector clients in Ghana.

Academic Service — Reviewer, SACAIR 2026 Program Committee

References

Prof. Ebenezer Owusu (PhD Supervisor)
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Prof. Jamal-Deen Abdulai (Head of Department)
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